

# NON-MF GROUPS

SAUERS

Some groups are not MF. The obstruction is a property-(T) subgroup conjugated into a proper subgroup of itself, together with a normal property-(T) subgroup of the resulting commutator defect. The unit group of the binary Leavitt algebra  $L_{\mathbb{F}_2}(1, 2)$  is a finitely generated simple property-(T) group with no nontrivial homomorphism to an MF group, and its reduced  $C^*$ -algebra is separable and stably finite but not MF. We also construct a sofic group that is not MF, whose canonical trace on the maximal group  $C^*$ -algebra is amenable but not quasidiagonal, and a torsion-free group with no nontrivial homomorphism to an MF group.

*Draft. This manuscript is under revision.*

## 1. INTRODUCTION

A countable group  $G$  is *MF* [CDE] if there are positive integers  $d_n$  and maps  $V_n: G \rightarrow \mathcal{U}(d_n)$ , with  $V_n(1) = 1$ , such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \quad (g \in G \setminus \{1\}).$$

For countable  $G$ , the second condition may equivalently be required along the full sequence with a constant independent of  $g$  [Kor, Propositions 2 and 7]. The strong convergence convention of [GKM, Sch] also requires the models to reproduce the operator norms of the left regular representation, so a group that is not MF as defined here is not MF in that convention either. Equivalently,  $G$  embeds in the unitary group of the norm matrix corona

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here and below, products of  $C^*$ -algebras mean bounded products, and  $\bigoplus$  denotes the ideal of norm-null sequences. We call a homomorphism  $G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$  a *corona homomorphism*. A separable  $C^*$ -algebra is MF if it embeds in a norm matrix corona [BK]. Write

$$\text{Rad}_{\text{MF}}(G) = \bigcap \{\ker f : f: G \rightarrow M, M \text{ an MF group}\}$$

for the *MF radical* of  $G$ . For countable  $G$  the quotient  $G/\text{Rad}_{\text{MF}}(G)$  is again MF, and hence the largest MF quotient of  $G$ : enumerate its nontrivial elements and separate each by an MF model, and at stage  $n$  take the direct sum of the first  $n$  models, each

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2020 *Mathematics Subject Classification*. Primary 46L05; Secondary 20F65, 22D10, 16S50.

*Key words and phrases*. MF groups, norm matrix coronas, property (T), asymptotic representations, Leavitt algebra.

carried far enough along its own sequence to have multiplicative defect below  $1/n$  on the first  $n$  elements while still separating its own element. This family is asymptotically multiplicative and separates every nontrivial element.

Some groups admit no such approximation. We construct groups for which every homomorphism to an MF group is trivial, as well as a sofic group with a nontrivial element killed by every MF homomorphism.

Our first counterexample is the group

$$W = \text{Cl}(X) \rtimes V,$$

where  $V$  is the ascending HNN extension of  $\mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$  along  $(v, A) \mapsto (2v, A)$ , the set  $X$  is the coset space of  $\mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$  in  $V$ , and  $\text{Cl}(X)$  is generated by involutions  $c_x$ ,  $x \in X$ , and a central involution  $\varepsilon$ , with  $[c_x, c_y] = \varepsilon$  for  $x \neq y$ . Every homomorphism from  $W$  to an MF group kills  $\varepsilon$  (Section 4).<sup>1</sup> This paper focuses on a different counterexample,

$$H = L_{\mathbb{F}_2}(1, 2)^\times,$$

the unit group of the binary Leavitt algebra, because  $H$  is simple and every homomorphism from  $H$  to an MF group is trivial. The right module  $R_R$  over  $R = L_{\mathbb{F}_2}(1, 2)$  satisfies  $R_R \cong R_R \oplus R_R$ , and  $H$  is its automorphism group; Khanh–Thanh show that  $H \cong \text{GL}_n(R) = \text{EL}_n(R)$  for every  $n \geq 2$  [K–T, Proposition 4.2 and Corollary 4.4]. OpenAI used this group to construct the first nonsolic group [OAI, Chapter 3].

We use the group commutator convention  $[g, h] = ghg^{-1}h^{-1}$ . The groups here fail to be MF because they contain a property-(T) subgroup that is conjugated into a proper subgroup of itself, together with an element that commutes with the subgroup while its conjugate does not. Call  $u \in G$  a *compressor* of  $L \leq G$  when  $uLu^{-1} \leq L$ . An element  $c$  of the centralizer  $C_G(L)$  commutes with  $L$ , so  $ucu^{-1}$  commutes with  $uLu^{-1}$  but need not commute with the rest of  $L$ . The normal subgroup of  $G$  generated by the commutators of all such elements  $ucu^{-1}$  with elements of  $L$  is

$$(1) \quad \mathfrak{D}_G(L) = \langle [ucu^{-1}, \ell] : uLu^{-1} \leq L, c \in C_G(L), \ell \in L \rangle_G.$$

For every homomorphism  $f: G \rightarrow G'$ ,

$$(2) \quad f(\mathfrak{D}_G(L)) \leq \mathfrak{D}_{G'}(f(L)),$$

since compressors, centralizing elements, and commutators of  $L$  map to compressors, centralizing elements, and commutators of  $f(L)$ , and conjugates map to conjugates.

The criterion behind every example is Theorem 2.8: for a property-(T) subgroup  $L \leq G$ , every normal property-(T) subgroup of  $G$  inside  $\mathfrak{D}_G(L)$  dies in every MF image.

The proof has two steps. First, the conjugation maps  $\text{Ad}(V_n(g))$  of an operator norm asymptotic representation form a homomorphism into a norm matrix corona, in which the Kazhdan projection  $P$  of  $L$  and the class  $U$  of  $(\text{Ad}(V_n(u)))$  satisfy  $U^*PU \leq P$ ; stable finiteness of the corona forces equality, so conjugation by  $u$  preserves the Hilbert–Schmidt asymptotic commutant of  $L$ , and every element of  $\mathfrak{D}_G(L)$  tends to 1 in normalized Hilbert–Schmidt norm (Corollary 2.5). Second, for a normal property-(T)

<sup>1</sup>The Lean proof was registered at Palomar on August 24, 2026: <https://palomar-registry.org/entry?id=PALOMAR-2026-08-24-000006&version=1>.

subgroup  $K \leq \mathfrak{D}_G(L)$ , a corona homomorphism nontrivial on  $K$  compresses to a corner, renormalized by its own rank, on which the Kazhdan projection of  $K$  vanishes while the limiting trace is the trivial character of  $K$ , which takes the value 1 on that projection (Theorem 2.7).<sup>2</sup>

The stable letter of an ascending HNN extension of a property-(T) group along a proper self-embedding is a compressor. For a unital ring  $R$ , write  $e_{ij}(a) = 1 + aE_{ij}$  and let  $\text{EL}_n(R)$  be the subgroup of  $\text{GL}_n(R)$  generated by the elements  $e_{ij}(a)$ ,  $i \neq j$ . A pair  $s, t \in R$  with  $ts = 1$  gives a compressor in the same way: an explicit element  $u \in \text{EL}_4(R)$  conjugates  $\text{EL}_3(R)$  into itself by  $e_{ij}(a) \mapsto e_{ij}(sat)$ , and if  $1 - st$  generates  $R$  as a two-sided ideal, then one commutator  $[ucu^{-1}, \ell]$  normally generates all of  $\text{EL}_4(R)$ . This happens once, in the universal unital associative ring  $\mathcal{C} = \mathbb{Z}\langle s_0, s_1, t_0, t_1 \rangle / (t_i s_j - \delta_{ij} : i, j \in \{0, 1\})$ , and  $\text{EL}_4(\mathcal{C})$  then maps to every  $\text{EL}_n(R)$  below with normally generating image.

Theorem 3.4 carries this to every countable unital ring with such a pair  $s, t$  and to every  $n \geq 2$ : every homomorphism from  $\text{EL}_n(R)$  to an MF group is trivial.

Let  $L_{\mathbb{F}_2}(1, 2)$  denote the binary Leavitt algebra [Lev], the unital  $\mathbb{F}_2$ -algebra generated by  $s_0, s_1, t_0, t_1$  subject to

$$(3) \quad t_i s_j = \delta_{ij}, \quad s_0 t_0 + s_1 t_1 = 1.$$

**Theorem A** (the unit group of the binary Leavitt algebra). *Let  $R = L_{\mathbb{F}_2}(1, 2)$  and let  $H = R^\times$  be its unit group. Then  $H \cong \text{EL}_4(R)$ , and  $H$  is finitely generated, nontrivial, simple, has property (T), and every homomorphism from  $H$  to an MF group is trivial. So  $H$  is not MF, and  $C_r^*(H)$  is separable and stably finite but not MF, while  $C_{\max}^*(H)$  is not finite: it contains a proper isometry.*

For a countable purely infinite simple ring  $R$  and  $n \geq 1$ , Theorem 3.8 identifies  $\text{Rad}_{\text{MF}}(\text{GL}_n(R))$  with the commutator subgroup, and the largest MF quotient of  $\text{GL}_n(R)$  with  $K_1(R)$ . For the unit group of  $L_k(1, d)$  the quotient is  $k^\times / (k^\times)^{d-1}$ .

**Theorem B** (an amenable nonquasidiagonal trace). *There is a sofic group  $W = W_0 \rtimes \mathbb{Z}$ , with  $W_0$  locally residually finite, that is not MF. The canonical trace on  $C_{\max}^*(W_0)$  is quasidiagonal, and the canonical trace on  $C_{\max}^*(W)$  is amenable and not quasidiagonal.*

Section 4 constructs  $W$  from the affine group  $\mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$  and Clifford lamps. The group  $W$  is sofic because locally residually finite groups are sofic and soficity passes to extensions with amenable quotient [ES, Theorem 1]. The trace answers the question of Brown [Br, discussion preceding Proposition 3.5.1] and of Schafhauser, Tikuisis and White [STW, Problem X(1)] whether every amenable trace is quasidiagonal. Tikuisis, White, and Winter proved that faithful traces on separable nuclear  $C^*$ -algebras satisfying the universal coefficient theorem are quasidiagonal [TWW]; here  $C_{\max}^*(W)$  is not nuclear, since  $W$  is not amenable. The canonical trace of  $C_{\max}^*(W_0)$  is quasidiagonal, so the crossed product by  $\mathbb{Z}$  destroys quasidiagonality of the canonical trace. Since  $W_0$  is a direct limit of residually finite groups, it is MF [Kor, Corollary 10

<sup>2</sup>A Hilbert–Schmidt bound on the multiplicative defect does not give an operator norm bound on the conjugation maps, so this argument does not apply to sofic or hyperlinear approximations.

and Proposition 13]. So MF groups are not closed under semidirect products with  $\mathbb{Z}$ , which answers a question of Korchagin [Kor, remark following Proposition 12].

Every example above has torsion, and in the lamp construction the obstruction is itself a torsion element. The last construction has none.

**Theorem C** (a torsion-free example). *There is a two-generated, finitely presented, torsion-free, acylindrically hyperbolic group  $Q$  with property (T) such that every homomorphism from  $Q$  to an MF group is trivial. In particular, no nontrivial quotient of  $Q$  is MF.*

Section 5 builds  $Q$  from Fournier-Facio’s torsion-free property-(T) group [FF, §2] and Hull’s small cancellation theorem [Hu]. Its reduced  $C^*$ -algebra is simple, has a unique tracial state and stable rank one, and is not MF.

**Related work.** Blackadar and Kirchberg introduced the notions of MF and NF algebras in a 1994 Oberwolfach abstract [BK94]. They wrote that the classes of NF, strong NF, and separable nuclear stably finite  $C^*$ -algebras might coincide. Their 1997 paper developed these notions and proved that a separable  $C^*$ -algebra is NF if and only if it is nuclear and MF [BK]. It also asked whether every separable nuclear stably finite  $C^*$ -algebra is quasidiagonal; for algebras that satisfy the universal coefficient theorem and have a faithful trace, the theorem of Tikuisis, White, and Winter recalled above answers this [TWW]. The question without nuclearity, whether every stably finite  $C^*$ -algebra is MF, is the MF problem, also posed by Blackadar and Kirchberg [GH, §6]. A positive answer would make every countable group MF, because the reduced group  $C^*$ -algebra is separable and stably finite; Lubotzky–Oppenheim and, separately, Thom noted this when they listed operator norm approximation of groups among the open approximation problems [LO, Th]. Carrión, Dadarlat, and Eckhardt introduced MF groups [CDE], and whether every countable group is MF remained open [Kor, BDL]. The MF problem itself was answered negatively in 2020: the failure of the Connes embedding problem [JNVWY] yields a  $\text{II}_1$  factor that embeds in no norm ultraproduct of matrix algebras, hence a separable stably finite  $C^*$ -algebra that is not MF [GH, Proposition 6.1 and Remark 6.2]. Theorem A gives a counterexample among reduced group  $C^*$ -algebras and answers the group question.

Theorem 2.8 uses the configuration of the nonsofic construction [OAI, Proposition 2.3]: a property-(T) subgroup  $L$  conjugated into itself by an element  $u$ , and an element  $c$  commuting with  $L$  whose conjugate  $ucu^{-1}$  does not. There, soficity and the rigidity theorems of Kun [Ku16] and Kun–Thom [KT19] force the centralizing subgroup to be locally embeddable into finite groups, and a copy of Thompson’s group  $V$  gives the contradiction. Here the rigidity comes from the Kazhdan projection of  $L$  in a norm matrix corona, which commutes with the compressor because the corona is stably finite, and the contradiction is a commutator  $[ucu^{-1}, \ell] \neq 1$  that every operator norm asymptotic representation sends to 1 in Hilbert–Schmidt norm. The group  $H$  is the group of [OAI], and Theorem C reuses Fournier-Facio’s configuration [FF]. Bachner–Dogon–Lubotzky showed that for groups of Deligne type, operator–Hilbert–Schmidt stability would imply that the group is not MF [BDL, Proposition 1.5]. Here the compression relation replaces the stability hypothesis.

## 2. ONE-SIDED COMPRESSION

## 2.1. Corona homomorphisms.

**Lemma 2.1.** *Let  $G$  be countable and  $K \leq G$ . Every homomorphism from  $G$  to an MF group is trivial on  $K$  if and only if every corona homomorphism from  $G$  is trivial on  $K$ . Let  $A$  be a unital  $C^*$ -algebra and let  $\pi: G \rightarrow \mathcal{U}(A)$  be an injective group homomorphism. If  $A$  embeds in a norm matrix corona, then  $G$  is MF. In particular, this applies to the canonical group homomorphisms into  $\mathcal{U}(C_{\max}^*(G))$  and  $\mathcal{U}(C_r^*(G))$ .*

*Proof.* The image of a corona homomorphism is a countable MF group. Conversely, compose a homomorphism to an MF group with a corona embedding of its image; the resulting corona homomorphism has the same kernel. If  $\iota$  is an embedding as in the last assertion, then  $g \mapsto \iota(\pi(g)) + 1 - \iota(1)$  is a corona homomorphism that is injective on  $G$ ; the complement term allows  $\iota$  to be nonunital.  $\square$

**2.2. Kazhdan transport in normalized Hilbert–Schmidt norm.** For an actual representation  $\rho: G \rightarrow \mathcal{U}(d)$  and  $u \in G$  with  $uLu^{-1} \leq L$ , the commutant  $\mathcal{C} = \rho(L)' \subseteq M_d(\mathbb{C})$  satisfies  $\rho(u)^*\mathcal{C}\rho(u) \subseteq \mathcal{C}$ , and the two spaces have the same finite dimension, so the inclusion is an equality; if  $c$  commutes with  $L$ , then  $\rho(ucu^{-1})$  commutes with  $\rho(L)$ , and  $\rho$  is trivial on  $\mathfrak{D}_G(L)$ . An asymptotic representation has no exact commutant to count. Property (T) supplies a Kazhdan projection in its place, and stable finiteness of the corona replaces the dimension count.

For  $a \in M_d(\mathbb{C})$  write

$$\|a\|_2 = \text{tr}_d(a^*a)^{1/2}, \quad \text{tr}_d = \frac{1}{d} \text{Tr}.$$

An *operator norm asymptotic representation* of a group  $G$  is a sequence of maps  $V_n: G \rightarrow \mathcal{U}(d_n)$  such that  $V_n(1) = 1$  and

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G).$$

The elements  $g$  with  $\|V_n(g) - 1\|_2 \rightarrow 0$  form a normal subgroup of  $G$ , because  $\|a\|_2 \leq \|a\|$ , the Hilbert–Schmidt norm is invariant under adjoints and unitary conjugation, and  $\|V_n(g^{-1}) - V_n(g)^*\| \rightarrow 0$ :

$$\|V_n(gh) - 1\|_2 \leq \|V_n(g) - 1\|_2 + \|V_n(h) - 1\|_2 + o(1),$$

$$\|V_n(g^{-1}) - 1\|_2 = \|V_n(g) - 1\|_2 + o(1), \quad \|V_n(ghg^{-1}) - 1\|_2 = \|V_n(h) - 1\|_2 + o(1).$$

For  $L \leq G$ , let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \left| \begin{array}{l} \sup_n \|x_n\|_2 < \infty, \\ \|V_n(\ell)x_n - x_n V_n(\ell)\|_2 \rightarrow 0 \quad (\ell \in L) \end{array} \right. \right\}$$

be the Hilbert–Schmidt asymptotic commutant of  $V(L)$ . Coordinatewise conjugation by  $V_n(g)$  defines a bijection, denoted  $\text{Ad}(V(g))$ , of the Hilbert–Schmidt bounded matrix sequences. A unital  $C^*$ -algebra is finite if each of its isometries is unitary, and stably finite if all matrix algebras over it are finite.

**Lemma 2.2.** *For every sequence of positive integers  $(m_n)$ , the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

*is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.*

*Proof.* Write  $A$  for the displayed corona. If  $v^*v = 1$  in  $A$  and  $(x_n)$  is a bounded lift of  $v$ , then  $\|x_n^*x_n - 1\| \rightarrow 0$ , so for all large  $n$  the matrix  $u_n = x_n(x_n^*x_n)^{-1/2}$  is unitary and  $\|u_n - x_n\| \rightarrow 0$ ; after assigning arbitrary unitary values to the finitely many remaining coordinates,  $(u_n)$  is a unitary lift of  $v$ , and  $vv^* = 1$ . So  $A$  is finite, and so is  $M_k(A)$ , which is the corona with matrix sizes  $km_n$ . Finally, if  $p \leq q$  are equivalent projections in a finite algebra  $B$  and  $w$  is a partial isometry with  $w^*w = q$  and  $ww^* = p$ , then  $w = qwq$ , so  $w + (1 - q)$  is an isometry of  $B$  and hence unitary; therefore  $p + (1 - q) = (w + (1 - q))(w + (1 - q))^* = 1$  and  $p = q$ .  $\square$

For a group  $L$  with property (T), the Kazhdan projection is the central projection  $e_L \in C_{\max}^*(L)$  whose image in every unitary representation of  $L$  is the orthogonal projection onto the  $L$ -invariant vectors [AW].

**Lemma 2.3** (one-sided order for the Kazhdan projection). *Let  $L$  have property (T), let  $B$  be a unital  $C^*$ -algebra, and let  $\pi: L \rightarrow \mathcal{U}(B)$  be a homomorphism. Let  $P \in B$  be the image of  $e_L$  under the  $*$ -homomorphism  $C_{\max}^*(L) \rightarrow B$  induced by  $\pi$ . If a unitary  $U \in B$  satisfies  $U\pi(L)U^* \subseteq \pi(L)$ , then  $U^*PU \leq P$ .*

*Proof.* Fix a faithful nondegenerate representation of  $B$  on a Hilbert space  $\mathcal{H}$ , and let  $\text{Fix } \pi(L)$  be the space of vectors fixed by  $\pi(L)$ . Then  $P$  is the orthogonal projection onto  $\text{Fix } \pi(L)$ . If  $\xi \in \text{Fix } \pi(L)$  and  $\ell \in L$ , then  $\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi$ , because  $U\pi(\ell)U^*$  belongs to  $\pi(L)$ . So  $U^*\text{Fix } \pi(L) \subseteq \text{Fix } \pi(L)$ . Since  $U^*PU$  is the orthogonal projection onto  $U^*\text{Fix } \pi(L)$ , it follows that  $U^*PU \leq P$ .  $\square$

**Theorem 2.4** (one-sided Kazhdan transport). *Let  $L \leq G$  have property (T), and let  $u \in G$  satisfy  $uLu^{-1} \leq L$ . Let  $(V_n)$  be an operator norm asymptotic representation of  $G$ . Then*

$$\text{Ad}(V(u))(\mathcal{C}_2(V, L)) = \mathcal{C}_2(V, L).$$

*Proof.* For unitaries  $A, B$ ,

$$\|\text{Ad}(A) - \text{Ad}(B)\| \leq 2\|A - B\|,$$

where the norm on the left is the operator norm on the Hilbert space  $(M_{d_n}(\mathbb{C}), \|\cdot\|_2)$ , so the maps  $\text{Ad}(V_n(g))$  are asymptotically multiplicative in operator norm and

$$\tilde{\sigma}(g) = [\text{Ad}(V_n(g))]_n \in \mathcal{U}(\mathcal{B}), \quad \mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) / \bigoplus_n B(M_{d_n}(\mathbb{C})),$$

is a homomorphism, where  $B(M_{d_n}(\mathbb{C}))$  denotes the operators on the Hilbert–Schmidt Hilbert space. The algebra  $\mathcal{B}$  is a norm matrix corona with coordinate sizes  $d_n^2$  after a choice of matrix units. Let  $P \in \mathcal{B}$  be the image of the Kazhdan projection of  $L$  [AW] under  $C_{\max}^*(L) \rightarrow \mathcal{B}$ , and lift  $P$  to orthogonal projections  $P_n$  on  $(M_{d_n}(\mathbb{C}), \|\cdot\|_2)$  by functional calculus.

A Hilbert–Schmidt bounded sequence  $(x_n)$  lies in  $\mathcal{C}_2(V, L)$  if and only if

$$\|P_n x_n - x_n\|_2 \longrightarrow 0.$$

In one direction, fix  $\varepsilon > 0$  and, by density of the group algebra in  $C_{\max}^*(L)$ , a finite set  $F \subseteq L$  and scalars  $(a_\ell)_{\ell \in F}$  with

$$\left\| \sum_{\ell \in F} a_\ell u_\ell - e_L \right\| < \varepsilon,$$

where  $u_\ell$  is the canonical unitary of  $\ell$ ; the trivial character, which sends  $e_L$  to 1, gives  $|\sum_{\ell \in F} a_\ell - 1| < \varepsilon$ . The element  $\sum_{\ell \in F} a_\ell \tilde{\sigma}(\ell)$  of  $\mathcal{B}$  is within  $\varepsilon$  of  $P$ , so

$$\limsup_n \left\| P_n - \sum_{\ell \in F} a_\ell \text{Ad}(V_n(\ell)) \right\| \leq \varepsilon.$$

If  $(x_n) \in \mathcal{C}_2(V, L)$  and  $\sup_n \|x_n\|_2 \leq c$ , then  $\text{Ad}(V_n(\ell))x_n = x_n + o(1)$  in  $\|\cdot\|_2$  for each  $\ell \in F$ , so

$$\limsup_n \|P_n x_n - x_n\|_2 \leq \varepsilon c + \left| \sum_{\ell \in F} a_\ell - 1 \right| c \leq 2\varepsilon c,$$

and  $\varepsilon$  was arbitrary. In the other direction,  $u_\ell e_L = e_L$  in  $C_{\max}^*(L)$ , so  $\tilde{\sigma}(\ell)P = P$  and  $\|\text{Ad}(V_n(\ell))P_n - P_n\| \rightarrow 0$ ; if  $\|P_n x_n - x_n\|_2 \rightarrow 0$ , then

$$\begin{aligned} \|\text{Ad}(V_n(\ell))x_n - x_n\|_2 &\leq 2\|x_n - P_n x_n\|_2 \\ &\quad + \|\text{Ad}(V_n(\ell))P_n - P_n\| \|x_n\|_2 \longrightarrow 0. \end{aligned}$$

Put  $U = \tilde{\sigma}(u)$ . Since  $uLu^{-1} \leq L$ ,  $U\tilde{\sigma}(L)U^* \subseteq \tilde{\sigma}(L)$ , so  $U^*PU \leq P$  by Lemma 2.3, and  $U^*PU = P$  by Lemma 2.2, since the two projections are unitarily equivalent. So  $UP = PU$ , and correspondingly  $\|\text{Ad}(V_n(u))P_n - P_n \text{Ad}(V_n(u))\| \rightarrow 0$ . For  $(x_n) \in \mathcal{C}_2(V, L)$ ,

$$\begin{aligned} \left\| P_n \text{Ad}(V_n(u))^{\pm 1} x_n - \text{Ad}(V_n(u))^{\pm 1} x_n \right\|_2 &\leq \left\| [P_n, \text{Ad}(V_n(u))^{\pm 1}] \right\| \|x_n\|_2 \\ &\quad + \|P_n x_n - x_n\|_2 \longrightarrow 0, \end{aligned}$$

so  $\text{Ad}(V(u))^{\pm 1}x \in \mathcal{C}_2(V, L)$  by the displayed equivalence.  $\square$

**Corollary 2.5.** *Let  $L \leq G$  have property (T) and let  $(V_n)$  be an operator norm asymptotic representation of  $G$ . Then*

$$\|V_n(d) - 1\|_2 \longrightarrow 0 \quad (d \in \mathfrak{D}_G(L)).$$

*Proof.* Let  $uLu^{-1} \leq L$ ,  $c \in C_G(L)$ , and  $\ell \in L$ , and write  $\mathcal{C}_2 = \mathcal{C}_2(V, L)$ . Since  $c$  commutes with  $L$ ,  $(V_n(c)) \in \mathcal{C}_2$ , so  $(V_n(u)V_n(c)V_n(u)^*) \in \mathcal{C}_2$  by Theorem 2.4, and then  $(V_n(ucu^{-1})) \in \mathcal{C}_2$  because  $\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\| \rightarrow 0$ . So  $\|V_n(\ell)V_n(ucu^{-1}) - V_n(ucu^{-1})V_n(\ell)\|_2 \rightarrow 0$ , and by asymptotic multiplicativity  $\|V_n([ucu^{-1}, \ell]) - 1\|_2 \rightarrow 0$ . The elements with this property form a normal subgroup of  $G$ , so it contains  $\mathfrak{D}_G(L)$ .  $\square$

**2.3. From Hilbert–Schmidt to operator norm.** Smallness in normalized Hilbert–Schmidt norm does not bound the operator norm: for

$$D_d = \text{diag}(-1, 1, \dots, 1) \in \mathcal{U}(d), \quad \|D_d - 1\|_2 = \frac{2}{\sqrt{d}} \rightarrow 0, \quad \|D_d - 1\| = 2.$$

A corner occupying a vanishing fraction of the matrix can therefore carry the entire obstruction, which is why the corner below is renormalized by its own rank.

Restricting a corona homomorphism  $\rho$  to a corner requires a correction: a projection  $q$  commuting with  $\rho(G)$  has lifts  $q_n$  that commute only asymptotically with unitary lifts  $U_n(g)$  of  $\rho(g)$ , so the compressions  $q_n U_n(g) q_n$  are only approximately unitary in the matrix corners.

**Lemma 2.6** (central corona corners). *Let  $\rho: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$  be a homomorphism from a countable group, and let  $q \in \mathcal{Q}_{\mathbf{d}}$  be a nonzero projection commuting with  $\rho(G)$ . Discard the coordinates on which a fixed projection lift of  $q$  vanishes; infinitely many remain, and on them there are ranks  $r_n \geq 1$ , an identification of the corner  $q\mathcal{Q}_{\mathbf{d}}q$  with the norm matrix corona  $\mathcal{Q}_{\mathbf{r}}$ , and an operator norm asymptotic representation  $W_n: G \rightarrow \mathcal{U}(r_n)$  whose corona homomorphism is  $g \mapsto q\rho(g)$  under this identification.*

*Proof.* Lift  $q$  to projections  $q_n \in M_{d_n}(\mathbb{C})$  by functional calculus. Since  $q \neq 0$ , infinitely many  $q_n$  are nonzero; retain those coordinates and put  $r_n = \text{rank}(q_n)$ . Identifying each corner  $q_n M_{d_n}(\mathbb{C}) q_n$  with  $M_{r_n}(\mathbb{C})$  identifies the corner  $q\mathcal{Q}_{\mathbf{d}}q$  with the corona  $\mathcal{Q}_{\mathbf{r}}$ : a bounded sequence  $(z_n)$  representing  $z \in \mathcal{Q}_{\mathbf{d}}$  gives the representative  $(q_n z_n q_n)$  of  $qzq$ . Since  $q$  commutes with  $\rho(G)$ , the map  $g \mapsto q\rho(g)$  is a homomorphism from  $G$  to the unitary group of the corner, with unit  $q$ . Each value is a unitary of  $\mathcal{Q}_{\mathbf{r}}$ , so, as in the proof of Lemma 2.2, it lifts to unitaries  $W_n(g) \in \mathcal{U}(r_n)$ , with  $W_n(1) = I_{r_n}$ . Multiplicativity in the corona makes  $(W_n)$  an operator norm asymptotic representation whose corona homomorphism is  $g \mapsto q\rho(g)$ .  $\square$

**Theorem 2.7** (normal Kazhdan subgroups). *Let  $G$  be countable and let  $K \trianglelefteq G$  have property (T). If every operator norm asymptotic representation  $(V_n)$  of  $G$  satisfies  $\|V_n(k) - 1\|_2 \rightarrow 0$  for all  $k \in K$ , then every corona homomorphism from  $G$  is trivial on  $K$ .*

*Proof.* Suppose that a corona homomorphism  $\Theta: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$  is nontrivial on  $K$ . Let  $e_K \in C_{\max}^*(K)$  be the Kazhdan projection of  $K$ , and let  $p \in \mathcal{Q}_{\mathbf{d}}$  be its image under the homomorphism  $C_{\max}^*(K) \rightarrow \mathcal{Q}_{\mathbf{d}}$  induced by  $\Theta|_K$ . In any faithful representation of the corona,  $p$  is the projection onto the  $K$ -fixed vectors. Since  $gKg^{-1} = K$ , the range of  $\Theta(g)p\Theta(g)^*$  is  $\Theta(g)\text{Fix } \Theta(K) = \text{Fix } \Theta(gKg^{-1}) = \text{Fix } \Theta(K)$ , so

$$\Theta(g)p\Theta(g)^* = p \quad (g \in G),$$

and  $q = 1 - p$  is nonzero because  $\Theta$  is nontrivial on  $K$ . By Lemma 2.6 there is an operator norm asymptotic representation  $W_n: G \rightarrow \mathcal{U}(r_n)$  whose corona homomorphism  $\hat{\Theta}$  is the coordinate restriction of  $g \mapsto q\Theta(g)$ . Let  $\pi: C_{\max}^*(K) \rightarrow \mathcal{Q}_{\mathbf{r}}$  be the homomorphism induced by  $\hat{\Theta}|_K$ . Since  $q$  commutes with  $\Theta(K)$ , the map  $a \mapsto q\Theta(a)$  is a homomorphism on  $C_{\max}^*(K)$  agreeing with  $\pi$  on  $K$ , so  $\pi(e_K)$  is the coordinate restriction of  $qp = 0$ .

Fix a free ultrafilter  $\omega$  and let  $\tau$  be the limit along  $\omega$  of the normalized traces of the coordinates  $M_{r_n}(\mathbb{C})$ . A norm-null sequence has vanishing traces, since  $|\text{tr}_{r_n}(x)| \leq \|x\|$ ,

so  $\tau$  is a well defined tracial state on  $\mathcal{Q}_r$ . The hypothesis applies to  $(W_n)$  itself, so

$$|\mathrm{tr}_{r_n}(W_n(k)) - 1| \leq \|W_n(k) - I_{r_n}\|_2 \longrightarrow 0 \quad (k \in K)$$

in the normalized Hilbert–Schmidt norm of  $M_{r_n}(\mathbb{C})$ , and therefore  $\tau(\pi(u_k)) = 1$  for every  $k \in K$ , where  $u_k$  is the canonical unitary of  $k$ . The trivial character  $\chi$  of  $K$  is the state of  $C_{\max}^*(K)$  with  $\chi(u_k) = 1$  for every  $k$ , and it has  $\chi(e_K) = 1$ . The two states  $\tau \circ \pi$  and  $\chi$  agree on every  $u_k$ , hence by linearity on the dense subalgebra  $\mathbb{C}[K]$ , hence on  $C_{\max}^*(K)$ . So

$$1 = \chi(e_K) = \tau(\pi(e_K)) = \tau(0) = 0,$$

a contradiction.  $\square$

**Theorem 2.8** (one-sided compression criterion). *Let  $G$  be countable and let  $L \leq G$  have property (T). If  $K \trianglelefteq G$  has property (T) and  $K \leq \mathfrak{D}_G(L)$ , then every homomorphism from  $G$  to an MF group is trivial on  $K$ . In particular, if such a  $K$  is nontrivial, then  $G$  is not MF, and if  $G$  has property (T) and  $\mathfrak{D}_G(L) = G$ , then every homomorphism from  $G$  to an MF group is trivial.*

*Proof.* By Corollary 2.5, every operator norm asymptotic representation of  $G$  satisfies  $\|V_n(k) - 1\|_2 \rightarrow 0$  for  $k \in K \leq \mathfrak{D}_G(L)$ . By Theorem 2.7, every corona homomorphism is then trivial on  $K$ , and by Lemma 2.1 so is every homomorphism to an MF group. If  $K \neq 1$ , then  $G$  does not embed in an MF group, so  $G$  is not MF. The last assertion is the case  $K = G = \mathfrak{D}_G(L)$ .  $\square$

#### 2.4. The maximal group $C^*$ -algebra.

**Proposition 2.9** (proper one-sided compression). *Let  $G$  be a countable group containing a property-(T) subgroup  $\Gamma$  and an element  $t$  with  $t\Gamma t^{-1} \subsetneq \Gamma$ . Then  $C_{\max}^*(G)$  contains a proper isometry. In particular,  $C_{\max}^*(G)$  is not stably finite, admits no faithful tracial state, and is neither residually finite-dimensional nor MF.*

*Proof.* Let  $P \in C_{\max}^*(G)$  be the image of the Kazhdan projection of  $\Gamma$  [AW] under the canonical map  $C_{\max}^*(\Gamma) \rightarrow C_{\max}^*(G)$ , and let  $u$  be the canonical unitary of  $t$ . Conjugation by  $u$  implements the isomorphism  $\Gamma \rightarrow t\Gamma t^{-1}$  on canonical unitaries, so by Lemma 2.3 applied with  $U = u$ ,  $P \leq uPu^*$ , where  $uPu^*$  is the image of the Kazhdan projection of  $t\Gamma t^{-1}$ . The inequality is strict: in the quasi-regular representation of  $G$  on  $\ell^2(G/t\Gamma t^{-1})$ , the point mass at the base coset is fixed by  $t\Gamma t^{-1}$  and moved by every element of  $\Gamma \setminus t\Gamma t^{-1}$ , so the two fixed-space projections differ. Put  $q = uPu^*$ , so that  $P < q$ ,  $Pq = qP = P$ ,  $uP = qu$ , and  $Pu^*(1 - q) = Pu^* - Pu^*uPu^* = 0$ , and put  $s = Pu^* + (1 - q)$ . Then

$$s^*s = uPu^* + (1 - q) = 1, \quad ss^* = P + (1 - q) = 1 - (q - P) \neq 1,$$

so  $s$  is a proper isometry, and  $\mathrm{diag}(s, 1, \dots, 1)$  is one in every matrix algebra over  $C_{\max}^*(G)$ , which is therefore not stably finite. A tracial state  $\tau$  has  $\tau(q - P) = \tau(s^*s) - \tau(ss^*) = 0$  with  $q - P$  a nonzero positive element, so no tracial state is faithful. MF algebras are stably finite, and a separable residually finite-dimensional  $C^*$ -algebra is MF [BK].  $\square$

A group can satisfy the hypothesis of Proposition 2.9 and be MF. The ascending HNN extension  $V$  of  $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$  along  $(v, A) \mapsto (2v, A)$ , used in Section 4, has the faithful matrix realization

$$V \cong \left\{ \begin{pmatrix} 2^k A & v \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SL}_3(\mathbb{Z}), v \in \mathbb{Z}[1/2]^3, k \in \mathbb{Z} \right\},$$

and reduction modulo odd integers separates its elements, so  $V$  is residually finite and MF [Kor, Corollary 10]. So the lamps of Section 4 are necessary for the non-MF conclusion there.

### 3. ONE-SIDED INVERSES AND ELEMENTARY GROUPS

The proof of Theorem 3.4 runs in rank four over the ring  $\mathcal{C}$ , with three coordinates for the subgroup  $\mathrm{EL}_3(\mathcal{C})$ , which has property (T) because  $\mathcal{C}$  is finitely generated, and one for its centralizer; two lemmas then carry the conclusion from  $\mathrm{EL}_4(\mathcal{C})$  to every  $\mathrm{EL}_n(R)$ . The compressor needs only  $ts = 1$ ; the ideal condition on  $1 - st$  enters in the last step of the rank-four argument and in the first lemma.

**Lemma 3.1** (two copies). *Let  $R$  be a unital ring with  $s, t \in R$  such that  $ts = 1$  and  $R(1 - st)R = R$ . Then there are  $v_0, v_1, w_0, w_1 \in R$  with  $w_i v_j = \delta_{ij}$ . Conversely, such elements satisfy  $w_0 v_0 = 1$  and  $w_1(1 - v_0 w_0)v_1 = 1$ .*

*Proof.* Put  $e = 1 - st$ , so that  $es = te = 0$ , and choose  $a_j, b_j \in R$ ,  $0 \leq j < m$ , with  $\sum_j a_j e b_j = 1$ . Put

$$v_0 = s^m, \quad w_0 = t^m, \quad v_1 = \sum_j s^j e b_j, \quad w_1 = \sum_j a_j e t^j.$$

For all  $i, j \geq 0$ ,  $et^i s^j e = \delta_{ij} e$ : for  $j > i$  the middle factor is  $s^{j-i}$ , which  $e$  kills on the left, and for  $i > j$  it is  $t^{i-j}$ , which  $e$  kills on the right. So  $w_0 v_0 = t^m s^m = 1$ ,  $w_1 v_1 = \sum_j a_j e b_j = 1$ ,  $w_0 v_1 = \sum_j t^{m-j} e b_j = 0$ , and  $w_1 v_0 = \sum_j a_j e s^{m-j} = 0$ . Conversely,  $w_1(1 - v_0 w_0)v_1 = w_1 v_1 - (w_1 v_0)(w_0 v_1) = 1$ .  $\square$

**Lemma 3.2** (normal generation in rank two). *Let  $R$  be a unital ring and let  $v, w, a, b \in R$  satisfy  $wv = 1$ ,  $ba = 1$ , and  $bv = 0$ . Then*

$$D = [e_{12}(v), e_{21}(b)] = \mathrm{diag}(1 + vb, 1)$$

*normally generates  $\mathrm{EL}_2(R)$ .*

*Proof.* The matrix  $e_{12}(v)e_{21}(b)e_{12}(-v)$  has rows  $(1 + vb, -vbw)$  and  $(b, 1 - bv)$ , so for  $bv = 0$  it is  $\begin{pmatrix} 1+vb & 0 \\ b & 1 \end{pmatrix}$ , and multiplying by  $e_{21}(-b)$  gives  $D = \mathrm{diag}(1 + vb, 1)$ . Let  $N$  be the normal closure of  $D$  in  $\mathrm{EL}_2(R)$ . For  $r \in R$ ,  $[D, e_{12}(ar)] = e_{12}((1 + vb)ar - ar) = e_{12}(vr)$ , since  $ba = 1$ ; so  $e_{12}(vR) \leq N$ . Put  $f = 1 - vw$ , an idempotent with  $fv = 0 = wf$ , and

$$z = vf + fw + 1 - f - vfw.$$

Put  $x = vf$ ,  $y = fw$ ,  $q = vfw$ , and  $c = 1 - f - q$ , so that  $z = x + y + c$ . From  $wv = 1$  and  $fv = wf = 0$ :  $x^2 = y^2 = 0$ ,  $xy = q$ ,  $yx = f$ , the idempotents  $f$  and  $q$

are orthogonal, and  $c$  annihilates  $x$  and  $y$  on both sides; so  $z^2 = q + f + c = 1$  and  $zf = xf + yf + cf = vf$ . The factorization

$$\text{diag}(z, z^{-1}) = e_{12}(z)e_{21}(-z^{-1})e_{12}(z)e_{12}(-1)e_{21}(1)e_{12}(-1)$$

puts  $h = \text{diag}(z, z)$  in  $\text{EL}_2(R)$ , and  $he_{12}(fr)h^{-1} = e_{12}(zfrz) = e_{12}(vfrz) \in N$ . So  $e_{12}(r) = e_{12}(vwr)e_{12}(fr) \in N$  for every  $r \in R$ , and conjugating by  $e_{12}(1)e_{21}(-1)e_{12}(1)$  gives  $e_{21}(R) \leq N$ .  $\square$

**Lemma 3.3** (a rank-four compression cell). *Let  $R$  be a unital ring, let  $s, t \in R$  satisfy  $ts = 1$ , and put  $e = 1 - st$ . In  $G = \text{EL}_4(R)$  let  $L = \text{EL}_3(R)$  occupy coordinates  $1, 2, 3$ . There are  $u, c \in G$  with  $uLu^{-1} \leq L$  and  $c \in C_G(L)$  such that*

$$ucu^{-1} = e_{12}(e), \quad [ucu^{-1}, e_{23}(1)] = e_{13}(e).$$

*Every off-diagonal entry of a matrix in  $uLu^{-1}$  has the form  $sat$  with  $a \in R$ .*

*Proof.* Here  $e^2 = e$  and  $es = te = 0$ . For  $i = 1, 2, 3$ , set

$$u_i = e_{4i}(t-1)e_{i4}(1)e_{4i}(s-1)e_{i4}(-t).$$

Its block on coordinates  $(i, 4)$  is  $\begin{pmatrix} s & e \\ 0 & t \end{pmatrix}$ , so  $u = u_3u_2u_1 \in \text{EL}_4(R)$  is the matrix

$$u = \begin{pmatrix} s & 0 & 0 & e \\ 0 & s & 0 & et \\ 0 & 0 & s & et^2 \\ 0 & 0 & 0 & t^3 \end{pmatrix},$$

invertible as a product of elementary matrices. For  $1 \leq i \neq j \leq 3$  and  $a \in R$ ,

$$(4) \quad ue_{ij}(a) = e_{ij}(sat)u :$$

both sides differ from  $u$  by one entry,  $sa$  in position  $(i, j)$ , since on the right the increment  $sat \cdot (s, \dots, et^{j-1})$  from row  $j$  of  $u$  contributes  $sats = sa$  in position  $(i, j)$  and  $sate t^{j-1} = 0$  in position  $(i, 4)$ . So  $uLu^{-1} \leq L$ , and (4) gives the last assertion.

The element

$$c = [e_{41}(e), e_{14}(t)] = \text{diag}(1, 1, 1, 1 + et)$$

is computed in the block on coordinates  $(1, 4)$  using  $te = 0$ , and a diagonal matrix of this shape commutes with every  $\text{diag}(A, 1)$ , so  $c \in C_G(L)$ . Both  $uc$  and  $e_{12}(e)u$  equal  $u + etE_{14}$ : the last column of  $uc$  is  $(e, et, et^2, t^3)^T(1 + et)$  and  $e^2 = e$ ,  $te = 0$ , while  $e_{12}(e)u$  adds  $e$  times the second row  $(0, s, 0, et)$  of  $u$  to its first row, and  $es = 0$ . So  $ucu^{-1} = e_{12}(e)$ , and the Steinberg relation  $[e_{12}(e), e_{23}(1)] = e_{13}(e)$  gives the second identity.  $\square$

**Theorem 3.4** (full complementary idempotents). *Let  $R$  be a countable unital associative ring. Suppose that  $s, t \in R$  satisfy*

$$ts = 1, \quad R(1 - st)R = R;$$

*that is,  $1 - st$  is a full idempotent, so that  $1 = \sum_j a_j(1 - st)b_j$  for finitely many  $a_j, b_j \in R$ . For every  $n \geq 2$ , every homomorphism from  $\text{EL}_n(R)$  to an MF group is trivial. Every homomorphism from  $\text{EL}_4(C)$  to an MF group is trivial; that group is finitely generated and has property (T); and there is a homomorphism from it to  $\text{EL}_n(R)$  whose image normally generates  $\text{EL}_n(R)$ .*

*Proof.* *The universal group.* In  $\mathcal{C}$  put  $s = s_0$ ,  $t = t_0$ , and  $e = 1 - st$ . Then  $ts = 1$ ,  $e^2 = e$ ,  $es = te = 0$ , and  $t_1es_1 = 1$ . Put  $G = \text{EL}_4(\mathcal{C})$  and  $L = \text{EL}_3(\mathcal{C})$  on coordinates 1, 2, 3. The ring  $\mathcal{C}$  is nonzero, since it maps onto  $L_{\mathbb{F}_2}(1, 2)$ , and finitely generated, so both groups have property (T) by [EJZ, Theorem 1.1], and  $\text{EL}_4(\mathcal{C})$  is finitely generated [BHV, Theorem 1.3.1].

By Lemma 3.3 there are  $u, c \in G$  with  $uLu^{-1} \leq L$ ,  $c \in C_G(L)$ , and

$$d = [ucu^{-1}, e_{23}(1)] = e_{13}(e) \in \mathfrak{D}_G(L).$$

Let  $N$  be the normal closure of  $d$  in  $G$ . For arbitrary  $a, b \in \mathcal{C}$ , the Steinberg relations give

$$[e_{41}(a), e_{13}(e)] = e_{43}(ae), \quad [e_{43}(ae), e_{32}(b)] = e_{42}(aeb).$$

So  $e_{42}(1) = e_{42}(t_1es_1) \in N$ . Elementary signed permutation matrices conjugate this element to every off-diagonal position, up to a sign, so  $e_{ij}(1) \in N$  whenever  $i \neq j$ . For distinct  $i, j, k$  and every  $r \in \mathcal{C}$ ,

$$[e_{ij}(1), e_{jk}(r)] = e_{ik}(r) \in N.$$

So  $N = G$ , and  $\mathfrak{D}_G(L) = G$ . By Theorem 2.8 with  $K = G$ , every homomorphism from  $G$  to an MF group is trivial.

*Rank two.* Now let  $R$  satisfy the hypothesis. By Lemma 3.1 choose  $v_0, v_1, w_0, w_1 \in R$  with  $w_iv_j = \delta_{ij}$ . Then  $\varphi: s_i \mapsto v_i, t_i \mapsto w_i$  is a unital ring homomorphism  $\mathcal{C} \rightarrow R$ , and it induces a homomorphism  $\text{EL}_4(\mathcal{C}) \rightarrow \text{EL}_4(R)$ . Put

$$\begin{aligned} S_1 &= v_0v_0, & S_2 &= v_0v_1, & S_3 &= v_1v_0, & S_4 &= v_1v_1, \\ T_1 &= w_0w_0, & T_2 &= w_1w_0, & T_3 &= w_0w_1, & T_4 &= w_1w_1, \end{aligned}$$

so that  $T_iS_j = \delta_{ij}$ , and put  $p = \sum_i S_iT_i$ . For a  $4 \times 4$  matrix  $A$  over  $R$  put

$$j(A) = 1 - p + \sum_{i,j} S_i A_{ij} T_j \in R.$$

Since  $(1-p)S_i = 0$  and  $T_j(1-p) = 0$ , we have  $j(AB) = j(A)j(B)$  and  $j(I) = 1$ , so  $\Psi(A) = \text{diag}(j(A), 1)$  defines a homomorphism  $\text{GL}_4(R) \rightarrow \text{GL}_2(R)$ , and for  $i \neq j$  and  $r \in R$ ,

$$\Psi(e_{ij}(r)) = \text{diag}(1 + S_i r T_j, 1) = [e_{12}(S_i r), e_{21}(T_j)]$$

by the computation in Lemma 3.2, since  $T_jS_i = 0$ . So  $\Psi(\text{EL}_4(R)) \leq \text{EL}_2(R)$ . Let  $\rho$  be a homomorphism from  $\text{EL}_2(R)$  to an MF group. Composed with  $\text{EL}_4(\mathcal{C}) \rightarrow \text{EL}_4(R) \rightarrow \text{EL}_2(R)$  it is trivial, so  $\rho$  kills the image  $\Psi(e_{12}(1)) = \text{diag}(1 + S_1T_2, 1)$  of  $e_{12}(1) \in \text{EL}_4(\mathcal{C})$ , which normally generates  $\text{EL}_2(R)$  by Lemma 3.2 with  $v = S_1$ ,  $w = T_1$ ,  $a = S_2$ ,  $b = T_2$ . So  $\rho$  is trivial, and the image of  $\text{EL}_4(\mathcal{C})$  in  $\text{EL}_2(R)$  normally generates  $\text{EL}_2(R)$ .

*All ranks.* For  $n \geq 3$ , signed permutation matrices conjugate  $e_{12}(r)$  to  $e_{ij}(\pm r)$  for all  $i \neq j$ , so the copy of  $\text{EL}_2(R)$  on coordinates 1, 2 normally generates  $\text{EL}_n(R)$ . A homomorphism from  $\text{EL}_n(R)$  to an MF group is trivial on that copy, hence trivial, and the image of  $\text{EL}_4(\mathcal{C})$  in that copy normally generates  $\text{EL}_n(R)$ .  $\square$

The non-MF conclusion passes up from corners: if a nonzero idempotent  $p \in R$  is such that the ring  $pRp$ , with unit  $p$ , satisfies the hypothesis, then  $A \mapsto A + (1-p)I_n$  embeds  $\text{EL}_n(pRp)$  in  $\text{EL}_n(R)$ , and restricting models shows that MF passes to subgroups, so

$\text{EL}_n(R)$  is not MF for every  $n \geq 2$ . The full-defect hypothesis itself need not pass to  $R$ , nor need every homomorphism from  $\text{EL}_n(R)$  to an MF group be trivial.

**Corollary 3.5.** *If  $R$  is a countable simple unital ring that is not directly finite, then every homomorphism from  $\text{EL}_n(R)$  to an MF group is trivial for every  $n \geq 2$ . The same conclusion holds for  $R = L_k(1, m)$ , for every countable field  $k$  and every  $m \geq 2$ .*

*Proof.* In the first case choose  $ts = 1 \neq st$ ; then  $1 - st$  is a nonzero idempotent, and since  $R$  is simple it generates  $R$  as a two-sided ideal. In the second case, with generators  $s_1, \dots, s_m, t_1, \dots, t_m$  subject to  $t_i s_j = \delta_{ij}$  and  $\sum_i s_i t_i = 1$ , take  $s = s_1$  and  $t = t_1$ ; then  $1 - s_1 t_1 = \sum_{i \geq 2} s_i t_i$  and

$$t_2(1 - s_1 t_1)s_2 = 1.$$

Theorem 3.4 applies in both cases.  $\square$

**Corollary 3.6.** *Let  $R$  be a countable unital ring. If  $R$  is not directly finite, then  $C_{\max}^*(\text{EL}_n(R))$  contains a proper isometry for every  $n \geq 4$ . If  $R \neq 0$  satisfies the hypothesis of Theorem 3.4, then for every  $n \geq 2$  the algebra  $C_r^*(\text{EL}_n(R))$  is separable, stably finite, and not MF, and the unit group  $R^\times$  is not MF.*

*Proof.* Choose  $s, t \in R$  with  $ts = 1 \neq st$ , and let  $S$  be the unital subring they generate. Let  $u$  and  $L = \text{EL}_3(S)$  be as in Lemma 3.3 for  $S$ ; then  $L$  has property (T) [EJZ, Theorem 1.1], and the compression is strict, since  $e(\text{sat}) = 0$  while  $e \cdot 1 = e \neq 0$  puts  $e_{12}(1)$  in  $L \setminus uLu^{-1}$ . By Proposition 2.9 applied to  $L \leq \text{EL}_n(R)$  and  $u$ ,  $C_{\max}^*(\text{EL}_n(R))$  contains a proper isometry.

If  $R \neq 0$  satisfies the hypothesis of Theorem 3.4, then  $G = \text{EL}_n(R)$  is countable and nontrivial, and by Theorem 3.4 it is not MF. The algebra  $C_r^*(G)$  is separable, stably finite because its canonical trace is faithful, and not MF by Lemma 2.1. Finally, the map  $j$  from the proof of Theorem 3.4 is injective, since  $T_{ij}(A)S_j = A_{ij}$ , and sends  $\text{EL}_4(R)$  into  $R^\times$ ; that group is not MF, and MF passes to subgroups. So  $R^\times$  is not MF.  $\square$

**Proposition 3.7** (torsion complementary idempotents). *Let  $R$  be a countable unital ring, let  $s, t \in R$  satisfy  $ts = 1$ , and suppose that  $e = 1 - st$  has finite additive order. Then for every  $n \geq 4$ ,*

$$\text{EL}_n(R, \text{Re}R) \leq \text{Rad}_{\text{MF}}(\text{EL}_n(R)),$$

where  $\text{EL}_n(R, \text{Re}R)$  is the normal closure of the  $e_{ij}(a)$  with  $a \in \text{Re}R$ . In particular  $\text{EL}_n(R)$  is not MF when  $e \neq 0$ , and if  $R$  has positive characteristic, then  $\text{EL}_4(R)$  MF implies that  $R$  is directly finite.

*Proof.* Let  $S$  be the unital subring generated by  $s, t$ , and let  $L = \text{EL}_3(S)$  occupy coordinates 1, 2, 3 of  $\text{EL}_4(S)$ ; it has property (T) [EJZ, Theorem 1.1]. Put  $f_{ij} = s^i e t^j$ . Since  $es = te = 0$ ,  $f_{ij} f_{kl} = \delta_{jk} f_{il}$ , so  $f_{ij}^2 = 0$  for  $i \neq j$ . Write  $D(a) = \text{diag}(1, 1, 1, a)$ ; for  $i \neq j$ ,  $D(1 \pm f_{ij}) = [e_{41}(f_{ii}), e_{14}(\pm f_{ij})]$  lies in  $\text{EL}_4(S)$ . Let  $u$  be the compressor of Lemma 3.3, put  $w_{ij} = (1 + f_{ji})(1 - f_{ij})(1 + f_{ji})$ ,  $r = w_{14}w_{25}$ , and  $v = uD(r)$ . Since  $D(r)$  centralizes  $L$ ,  $vLv^{-1} \leq L$ . Put

$$c = D(1 + f_{02}), \quad y = e_{42}(f_{10}), \quad z = D(1 + f_{12}), \quad \ell = e_{21}(1), \quad B = \langle L, v, c, y \rangle.$$

Then  $c \in C_B(L)$ , and from  $rf_{02}r^{-1} = f_{05}$ ,  $rf_{12}r^{-1} = f_{45}$ ,  $f_{02}t^3 = f_{05}$ , and  $t^3f_{45} = f_{12}t^3$ , direct multiplication with the displayed  $u$  gives  $vcv^{-1} = e_{14}(f_{02})$  and  $vzv^{-1} = z$ . So  $d = [vcv^{-1}, \ell] = e_{24}(-f_{02})$  lies in  $\mathfrak{D}_B(L)$ , and  $[y, d] = D(1 - f_{12}) = z^{-1}$  since  $f_{02}f_{10} = 0$  and  $f_{10}f_{02} = f_{12}$ ; hence  $z \in \mathfrak{D}_B(L)$ . The element  $z$  commutes with  $L$  and  $v$ , and with  $c$  and  $y$  because  $f_{12}f_{02} = f_{02}f_{12} = f_{12}f_{10} = 0$ , so  $\langle z \rangle$  is central in  $B$ . If  $me = 0$  then  $z^m = D(1 + mf_{12}) = 1$ , and  $z \neq 1$  because  $tf_{12}s^2 = e \neq 0$ . So  $\langle z \rangle$  is a nontrivial finite central subgroup of  $B$  inside  $\mathfrak{D}_B(L)$ , and Theorem 2.8 applied to  $B$  kills  $z$  under every homomorphism from  $B$ , hence from  $\text{EL}_n(R)$ , to an MF group. Then

$$[z, e_{41}(f_{20})] = e_{41}(f_{10}), \quad [e_{24}(t), e_{41}(f_{10})] = e_{21}(e)$$

are killed, hence every  $e_{ij}(aeb)$  by the Steinberg relations and signed permutations, hence  $\text{EL}_n(R, ReR)$ . For  $e \neq 0$ ,  $e_{21}(e) \neq 1$ . If  $R$  has characteristic  $p > 0$  then  $pe = 0$  for every such  $e$ , and the contrapositive gives direct finiteness.  $\square$

The Toeplitz–Jacobson algebra  $J = \mathbb{F}_2\langle s, t \mid ts = 1 \rangle$  has  $1 - st \neq 0$  of order two, while  $J/J(1 - st)J \cong \mathbb{F}_2[z, z^{-1}] \neq 0$ , so  $1 - st$  is not full and Theorem 3.4 does not apply; Proposition 3.7 shows that  $\text{EL}_n(J)$  is not MF for  $n \geq 4$ . The two results are complementary: fullness gives trivial MF images at every rank  $n \geq 2$ , torsion of the defect gives non-MF at rank four with no fullness.

**3.1. The binary example.** Put  $R = L_{\mathbb{F}_2}(1, 2)$  and  $H = R^\times$ . By (3), the maps  $x \mapsto (t_0x, t_1x)$  and  $(y, z) \mapsto s_0y + s_1z$  are mutually inverse isomorphisms of right  $R$ -modules between  $R$  and  $R \oplus R$ , and  $H \cong \text{GL}_4(R) = \text{EL}_4(R)$  [K–T, Proposition 4.2 and Corollary 4.4]. We identify  $H$  with  $\text{EL}_4(R)$ .

*Proof of Theorem A.* The relations give  $t_0s_0 = 1$  and  $t_1(1 - s_0t_0)s_1 = 1$ , so  $R$  satisfies the hypothesis of Theorem 3.4, and every homomorphism from  $H$  to an MF group is trivial. The ring  $R$  is finitely generated, so  $H$  has property (T) by [EJZ, Theorem 1.1], and hence is finitely generated [BHV, Theorem 1.3.1]. Since  $e_{12}(1) \neq 1$ , the group  $H$  is nontrivial.

Let  $N$  be a normal subgroup of  $H$ . The ring  $R$  is purely infinite simple [AA] and therefore an exchange ring [Ara], so by Preusser’s sandwich theorem [Pre, Theorem 3] there is an ideal  $I$  of  $R$  with

$$\text{EL}_4(R, I) \leq N \leq C_4(R, I),$$

where  $\text{EL}_4(R, I)$  is the normal closure in  $\text{EL}_4(R)$  of the elementary matrices  $e_{ij}(a)$  with  $a \in I$ , and  $C_4(R, I)$  is the preimage of the center of  $\text{GL}_4(R/I)$ . Since  $R$  is simple, either  $I = R$ , and then  $N \geq \text{EL}_4(R) = H$ , or  $I = 0$ , and then every  $g \in N$  is central in  $\text{GL}_4(R)$ . In the second case, commuting with each  $e_{ij}(1)$  forces  $g = \lambda I_4$  with  $\lambda \in R^\times$ , and commuting with each  $e_{ij}(a)$  then gives  $\lambda a = a\lambda$  for all  $a \in R$ . So  $\lambda$  lies in  $Z(R) = \mathbb{F}_2$  [AC, Corollary 4.3], and  $N = 1$ . So  $H$  is simple. It is nontrivial, and every homomorphism from it to an MF group is trivial, so  $H$  is not MF.

Since  $R \neq 0$  satisfies the hypothesis of Theorem 3.4, Corollary 3.6 shows that  $C_r^*(H)$  is separable, stably finite, and not MF, and that  $C_{\max}^*(H)$  contains a proper isometry.  $\square$

All rings in the following discussion are unital and associative. Two idempotents  $e, f$  of a ring are *equivalent* if  $e = xy$  and  $f = yx$  for some  $x, y$ ; an idempotent  $e$  is *infinite* if  $e = f + g$  for orthogonal idempotents  $f, g$  with  $f$  equivalent to  $e$  and  $g \neq 0$ ; and a simple ring is *purely infinite* if every nonzero right ideal contains an infinite idempotent [AGP, Definitions 1.2]. The algebras  $L_k(1, d)$  are purely infinite simple [AA]. Let  $\mathrm{GL}(R)$  and  $\mathrm{EL}(R)$  be the direct limits of  $\mathrm{GL}_n(R)$  and  $\mathrm{EL}_n(R)$  along  $A \mapsto \mathrm{diag}(A, 1)$ ; the subgroup  $\mathrm{EL}(R)$  is normal in  $\mathrm{GL}(R)$  by Whitehead's lemma, and  $K_1(R) = \mathrm{GL}(R)/\mathrm{EL}(R)$ . Write  $\kappa: R^\times \rightarrow K_1(R)$  for the canonical map.

**Theorem 3.8** (MF quotients of unit groups). *Let  $R$  be a countable unital purely infinite simple ring and let  $n \geq 1$ . Then every homomorphism from  $\mathrm{GL}_n(R)$  to an MF group factors uniquely through the canonical map  $\mathrm{GL}_n(R) \rightarrow K_1(R)$ , and  $K_1(R)$ , a countable abelian group, is MF. Equivalently,  $\mathrm{Rad}_{\mathrm{MF}}(\mathrm{GL}_n(R)) = [\mathrm{GL}_n(R), \mathrm{GL}_n(R)]$  and  $\mathrm{GL}_n(R)/\mathrm{Rad}_{\mathrm{MF}}(\mathrm{GL}_n(R)) \cong K_1(R)$ .*

*Proof.* The ring  $M_n(R)$  is again countable, purely infinite, and simple [AGP, Corollary 1.7], and  $K_1(M_n(R)) \cong K_1(R)$  by Morita invariance, so it suffices to treat  $n = 1$ . Write  $H = R^\times$  and  $N = \mathrm{Rad}_{\mathrm{MF}}(H)$ . Ara, Goodearl, and Pardo show that  $\kappa$  is surjective with kernel  $[H, H]$  [AGP, Theorem 2.4]. A countable abelian group  $A$  is MF:  $C_{\max}^*(A)$  is commutative and separable, hence residually finite-dimensional and MF [BK], and Lemma 2.1 applies. So  $\kappa$  is a homomorphism to an MF group, and  $N \leq \ker \kappa$ .

For the converse we use Theorem 3.4 in the following form. Every nonzero idempotent  $e$  of  $R$  is infinite [AGP, Proposition 1.5], so  $eRe$  contains  $s, t$  with  $ts = e \neq st$ , and  $eRe$  is simple, so  $e - st$  generates  $eRe$  as a two-sided ideal; hence every homomorphism from  $\mathrm{EL}_m(eRe)$ ,  $m \geq 2$ , to an MF group is trivial. Consequently, if  $e$  and  $P$  are nonzero idempotents and  $\theta: M_m(eRe) \rightarrow PRP$  is a ring isomorphism, then

$$(5) \quad 1 - P + \theta(A) \in N \quad (A \in \mathrm{EL}_m(eRe)),$$

since these units form the image of  $\mathrm{EL}_m(eRe)$  under a homomorphism to  $H$ .

We use two facts from the proof of [AGP, Theorem 2.4], which rest on [AGP, Proposition 1.5 and Corollary 1.7] and on the elementary reduction of Menal and Moncasi [MM, proof of Theorem 2.2, and the Remark after Corollary 2.3].

- (a) There are orthogonal idempotents  $e_1, \dots, e_m$  with sum 1,  $m \geq 4$ , such that  $e_1, \dots, e_{m-1}$  are pairwise equivalent and  $e_m$  is equivalent to an idempotent  $f \leq e_1$ . Then  $R$  is the ring of  $m \times m$  matrices over  $T = e_1 R e_1$ , with identity  $\mathrm{diag}(1_T, \dots, 1_T, f)$ , whose last column has entries in  $Tf$  and whose last row has entries in  $fT$ , and every unit  $u$  of  $R$  factors as  $u = gvh$  with  $g$  and  $h$  products of elementary matrices  $e_{ij}(x)$  of this matrix ring and  $v = e_1 + (1 - e_1)v(1 - e_1)$ .
- (b) For a nonzero idempotent  $e$  and  $n \geq 2$  there are an idempotent  $f < e$  equivalent to  $e$  and orthogonal idempotents  $r_2, \dots, r_n \leq e - f$ , each equivalent to 1, such that, with  $r_1 = 1 - e + f$  and  $P = r_1 + \dots + r_n$ , some ring isomorphism  $\theta: M_n(R) \rightarrow PRP$  sends  $\mathrm{diag}(v, 1, \dots, 1)$  to  $(1 - e)v(1 - e) + f + r_2 + \dots + r_n$  for every unit  $v = e + (1 - e)v(1 - e)$ .

Let  $u \in \ker \kappa$  and write  $u = gvh$  as in (a). The elementary matrices with  $i, j \leq m - 1$  form  $\mathrm{EL}_{m-1}(T)$  inside  $(1 - e_m)R(1 - e_m) \cong M_{m-1}(T)$ , so they lie in  $N$  by (5); for

distinct  $i, j \leq m-1$ ,

$$e_{im}(x) = [e_{ij}(1), e_{jm}(x)], \quad e_{mj}(x) = [e_{mi}(x), e_{ij}(1)]$$

are products of an element of  $\text{EL}_{m-1}(T)$  and a conjugate of its inverse, so they lie in the normal subgroup  $N$  as well. Hence  $u \equiv v$  modulo  $N$ , and  $\kappa(v) = \kappa(u) = 0$  because  $N \leq \ker \kappa$ .

So let  $v = e + (1-e)v(1-e)$  with  $e \neq 0$  and  $\kappa(v) = 0$ . By the definition of  $K_1$ ,  $\text{diag}(v, 1, \dots, 1) \in \text{EL}_n(R)$  for some  $n \geq 2$ . With  $\theta$  and  $P$  as in (b),  $v = 1 - P + \theta(\text{diag}(v, 1, \dots, 1))$ , which lies in  $N$  by (5). Hence  $\ker \kappa \leq N$ .  $\square$

**Corollary 3.9** (Leavitt unit groups). *Let  $k$  be a countable field, let  $d \geq 2$ , let  $R = L_k(1, d)$ , and let  $H = R^\times$ . Then  $H \cong \text{GL}_d(R)$ , and under this identification  $\text{Rad}_{\text{MF}}(H) = [H, H] = \text{EL}_d(R)$  and*

$$H/\text{EL}_d(R) \cong K_1(R) \cong k^\times / (k^\times)^{d-1}.$$

*Proof.* The maps  $x \mapsto (t_1x, \dots, t_dx)$  and  $(y_1, \dots, y_d) \mapsto \sum_i s_i y_i$  are mutually inverse isomorphisms of right  $R$ -modules between  $R$  and  $R^d$ , so  $R \cong M_d(R)$  and  $H \cong \text{GL}_d(R)$ . Theorem 3.8 identifies the radical with  $[H, H]$  and the quotient with  $K_1(R)$ . Khanh–Thanh show that  $\text{GL}_d(R) = \text{EL}_d(R)D_d(k)$ , where  $D_d(k)$  is the abelian group of diagonal matrices with entries in  $k^\times$ , which normalizes  $\text{EL}_d(R)$ , and that  $K_1(R) \cong k^\times / (k^\times)^{d-1}$  [K–T, proof of Theorem 7.2]. So  $[H, H] \leq \text{EL}_d(R)$ , while  $\text{EL}_d(R) \leq [H, H]$  because every homomorphism from  $\text{EL}_d(R)$  to an MF group is trivial (Theorem 3.4).  $\square$

For  $d = 2$  the quotient is trivial for every countable field  $k$ , so every homomorphism from  $L_k(1, 2)^\times$  to an MF group is trivial. For  $k = \mathbb{F}_q$  the quotient is cyclic of order  $\gcd(q-1, d-1)$ .

#### 4. AMENABLE TRACES AND CLIFFORD LAMPS

Brown observed that it was unknown whether every amenable trace on a  $C^*$ -algebra is quasidiagonal [Br, discussion preceding Proposition 3.5.1]. For a separable unital  $C^*$ -algebra  $A$  we use the sequential form of his definitions. A tracial state  $\tau$  on  $A$  is *amenable* if there are u.c.p. maps  $\phi_n: A \rightarrow M_{d_n}(\mathbb{C})$  such that

$$\|\phi_n(ab) - \phi_n(a)\phi_n(b)\|_2 \longrightarrow 0, \quad \text{tr}_{d_n}(\phi_n(a)) \longrightarrow \tau(a) \quad (a, b \in A).$$

It is *quasidiagonal* if the first limit holds in operator norm instead of normalized Hilbert–Schmidt norm, with the same trace convergence.

For a countable group  $G$ , let  $u_g$  denote the canonical unitary in  $C_{\max}^*(G)$ . The canonical trace  $\tau_G$  is determined by

$$\tau_G(u_g) = \begin{cases} 1, & g = 1, \\ 0, & g \neq 1. \end{cases}$$

**Theorem 4.1** (canonical-trace obstruction). *Let  $G$  be a countable group. If  $G$  is not MF, then its canonical trace  $\tau_G$  on  $C_{\max}^*(G)$  is not quasidiagonal. In particular, if  $\tau_G$  is amenable, then  $\tau_G$  is an amenable trace that is not quasidiagonal.*

*Proof.* Suppose that  $\tau_G$  is quasidiagonal, with u.c.p. maps  $\phi_n: C_{\max}^*(G) \rightarrow M_{d_n}(\mathbb{C})$  as in the definition. Asymptotic multiplicativity in operator norm makes  $a \mapsto [(\phi_n(a))_n]$  a unital  $*$ -homomorphism  $\Phi: C_{\max}^*(G) \rightarrow \mathcal{Q}_d$ . If  $\Phi(u_g) = 1$ , then  $\|\phi_n(u_g) - 1\| \rightarrow 0$ , so  $\text{tr}_{d_n}(\phi_n(u_g)) \rightarrow 1$ , while  $\text{tr}_{d_n}(\phi_n(u_g)) \rightarrow \tau_G(u_g)$ ; so  $g = 1$ . Hence  $g \mapsto \Phi(u_g)$  is a corona homomorphism injective on  $G$ , and  $G$  is MF, contrary to the hypothesis.  $\square$

A group  $N$  is *locally residually finite* if each of its finitely generated subgroups is residually finite.

**Proposition 4.2** (amenable extensions of locally residually finite groups). *Let  $1 \rightarrow N \rightarrow G \rightarrow A \rightarrow 1$  be an extension of countable groups with  $N$  locally residually finite and  $A$  amenable. Then the canonical trace of  $C_{\max}^*(N)$  is quasidiagonal, and the canonical trace of  $C_{\max}^*(G)$  is amenable.*

*Proof.* Write  $g \mapsto \bar{g}$  for the quotient map  $G \rightarrow A$ , and fix a finite set  $\Sigma \subseteq G$ , a finite Følner set  $F \subseteq A$ , and a section  $\sigma: A \rightarrow G$ . For  $g \in \Sigma$  and  $x \in F$  put

$$b(g, x) = \sigma(\bar{g}x)^{-1}g\sigma(x) \in N.$$

The subgroup  $N_0$  generated by these finitely many elements is residually finite; choose a finite quotient  $\theta: N_0 \rightarrow \Delta$  with  $\theta(b(g, x)) \neq 1$  whenever  $b(g, x) \neq 1$ , put  $\Lambda = \ker \theta$ , and choose representatives  $r_q \in N_0$  of the cosets of  $\Lambda$  in  $N_0$ . The cosets  $\sigma(x)r_q\Lambda$ ,  $x \in F$ ,  $q \in \Delta$ , are pairwise distinct, since equality forces  $x = x'$  in  $A$  and then  $q = q'$ ; let  $T$  be their set, let  $\Pi$  be the projection of  $\ell^2(G/\Lambda)$  onto  $\ell^2(T)$ , and put  $\Phi(b) = \Pi\lambda(b)\Pi|_{\ell^2(T)}$  for  $b \in C_{\max}^*(G)$ , where  $\lambda$  is the quasi-regular representation. This map is u.c.p., and for  $g \in \Sigma$  and  $\bar{g}x \in F$ ,

$$g\sigma(x)r_q\Lambda = \sigma(\bar{g}x)r_{\theta(b(g,x))q}\Lambda \in T.$$

So for  $g, h \in \Sigma$  the identity  $\Phi(u_{gh}) - \Phi(u_g)\Phi(u_h) = \Pi\lambda(g)(1 - \Pi)\lambda(h)\Pi$  and the rank of  $(1 - \Pi)\lambda(h)\Pi$ , at most  $|\{x \in F : \bar{h}x \notin F\}| |\Delta|$ , give

$$\|\Phi(u_{gh}) - \Phi(u_g)\Phi(u_h)\|_2^2 \leq \frac{|\{x \in F : \bar{h}x \notin F\}|}{|F|}.$$

The normalized trace of  $\Phi(u_g)$  is the fraction of points of  $T$  fixed by  $g$ , and for  $g \in \Sigma \setminus \{1\}$  it is zero: if  $\bar{g} \neq 1$ , then  $g$  moves the  $F$ -coordinate of every point, and if  $\bar{g} = 1$ , then  $g\sigma(x)r_q\Lambda = \sigma(x)r_{\theta(b(g,x))q}\Lambda$  with  $b(g, x) = \sigma(x)^{-1}g\sigma(x) \neq 1$ , so  $\theta(b(g, x))q \neq q$ . Taking Følner sets with  $|\{x \in F : \bar{h}x \notin F\}|/|F| \rightarrow 0$  for the finitely many  $h \in \Sigma$ , along an exhaustion of  $G$  by finite sets  $\Sigma$ , the maps  $\Phi$  are asymptotically multiplicative in normalized Hilbert–Schmidt norm and their traces converge to  $\tau_G$  on the canonical unitaries; density and contractivity extend both limits to  $C_{\max}^*(G)$ .

For the canonical trace of  $C_{\max}^*(N)$ , run the same construction with  $N$  in place of  $G$ ,  $A = 1$ , and  $F = \{1\}$ : no boundary term appears, so the maps  $\Phi$  are multiplicative on  $\Sigma$  in operator norm, and their traces vanish on  $\Sigma \setminus \{1\}$ ; the same limits give quasidiagonality.  $\square$

We now construct a group to which Theorem 2.8 applies through a finite central subgroup: its intrinsic defect subgroup contains a commutator whose square is a central

involution. Let  $\Gamma$  be a countable group with property (T), let  $\alpha: \Gamma \rightarrow \Gamma$  be injective but not surjective, and choose  $a \in \Gamma \setminus \alpha(\Gamma)$ . Put

$$T_\alpha = \varinjlim (\Gamma \xrightarrow{\alpha} \Gamma \xrightarrow{\alpha} \cdots), \quad V = T_\alpha \rtimes \langle t \rangle,$$

where  $tgt^{-1} = \alpha(g)$  on the level-zero copy of  $\Gamma$ , so that  $V$  is the ascending HNN extension of  $\Gamma$  along  $\alpha$  and  $T_\alpha = \bigcup_{n \geq 0} t^{-n} \Gamma t^n$ . Put  $X = V/\Gamma$ , the left-coset space. The Clifford lamp group  $\text{Cl}(X)$  is the group with generators  $\varepsilon$  and  $c_x$ ,  $x \in X$ , and relations

$$\varepsilon^2 = c_x^2 = 1, \quad [\varepsilon, c_x] = 1, \quad [c_x, c_y] = \varepsilon \quad (x \neq y).$$

To see that  $\varepsilon \neq 1$ , fix a total order on  $X$ , let  $\mathbb{F}_2^{(X)}$  be the  $\mathbb{F}_2$ -vector space of finitely supported functions  $X \rightarrow \mathbb{F}_2$ , and for  $f, g \in \mathbb{F}_2^{(X)}$  put  $B(f, g) = \sum_{x > y} f(x)g(y)$ . Since  $B$  is bilinear,

$$(a, f)(b, g) = (a + b + B(f, g), f + g)$$

is a group law on  $\mathbb{F}_2 \times \mathbb{F}_2^{(X)}$ , in which  $(1, 0)$  is a central involution, each  $(0, \delta_x)$  is an involution, and for  $x \neq y$  exactly one of  $B(\delta_x, \delta_y)$  and  $B(\delta_y, \delta_x)$  equals 1, so the commutator of  $(0, \delta_x)$  and  $(0, \delta_y)$  is  $(1, 0)$ . So  $\varepsilon \mapsto (1, 0)$ ,  $c_x \mapsto (0, \delta_x)$  defines a homomorphism  $\text{Cl}(X) \rightarrow \mathbb{F}_2 \times \mathbb{F}_2^{(X)}$ . The relations let us write every element of  $\text{Cl}(X)$  as

$$\varepsilon^a c_{x_1} \cdots c_{x_r}, \quad a \in \mathbb{F}_2, \quad x_1 < \cdots < x_r,$$

and this word maps to  $(a, \delta_{x_1} + \cdots + \delta_{x_r})$ , so the expression is unique and the homomorphism is an isomorphism. In particular, for a finite subset  $Y \subseteq X$ , the subgroup generated by  $\varepsilon$  and the  $c_y$  with  $y \in Y$  has order  $2^{|Y|+1}$ , so  $\text{Cl}(X)$  is countable and locally finite. Killing  $\varepsilon$  leaves  $\mathbb{F}_2^{(X)} \cong \bigoplus_X C_2$ , so  $\text{Cl}(X)$  is a central extension of  $\bigoplus_X C_2$  by  $C_2$ , and  $W$  below is a central extension of the permutational wreath product  $C_2 \wr_X V$  by  $\langle \varepsilon \rangle$ . The relations are invariant under permutations of  $X$ , so every permutation of  $X$  induces an automorphism of  $\text{Cl}(X)$  that permutes the  $c_x$  accordingly and fixes  $\varepsilon$ . In this way  $V$  acts on  $\text{Cl}(X)$  through its action on  $X$ , and we put

$$(6) \quad W = \text{Cl}(X) \rtimes V.$$

**Proposition 4.3** (Clifford obstruction from a proper self-embedding). *The group  $W$  is not MF. More precisely, every homomorphism from  $W$  to an MF group kills  $\varepsilon$ .*

*Proof.* Identify  $\Gamma$  with its level-zero copy in  $V \leq W$ , and let  $c$  be the lamp at the root coset  $\Gamma \in X$ ; it centralizes  $\Gamma$ , which fixes that coset. Then  $tct^{-1}$  is the lamp at  $t\Gamma$  and  $a(tct^{-1})a^{-1}$  is the lamp at  $at\Gamma$ . These cosets are distinct, because  $t\Gamma = at\Gamma$  would mean  $a \in t\Gamma t^{-1} = \alpha(\Gamma)$ ; so  $x = tct^{-1}$  and  $y = axa^{-1}$  are distinct lamps, involutions with  $[x, y] = \varepsilon \neq 1$ . Then  $d = [x, a] = xy$  and  $d^2 = xyxy = [x, y] = \varepsilon$ , and  $d = [tct^{-1}, a]$  is a generator of  $\mathfrak{D}_W(\Gamma)$ , since  $t\Gamma t^{-1} \leq \Gamma$ ,  $c \in C_W(\Gamma)$ , and  $a \in \Gamma$ . So  $\langle \varepsilon \rangle \leq \mathfrak{D}_W(\Gamma)$ . The subgroup  $\langle \varepsilon \rangle$  is central, so normal, and it is finite, so it has property (T). So Theorem 2.8 applies to the countable group  $W$  with  $L = \Gamma$  and  $K = \langle \varepsilon \rangle$ : every homomorphism from  $W$  to an MF group is trivial on  $\varepsilon$ . In particular,  $W$  is not MF.  $\square$

**Proposition 4.4** (the locally residually finite case). *Suppose in addition that  $\Gamma$  is residually finite and that  $[\Gamma : \alpha(\Gamma)] < \infty$ . Then*

$$W \cong W_0 \rtimes \mathbb{Z}, \quad W_0 = \text{Cl}(X) \rtimes T_\alpha,$$

where  $W_0$  is locally residually finite. Consequently  $W$  is sofic but not MF, the canonical trace on  $C_{\max}^*(W_0)$  is quasidiagonal, and the canonical trace on  $C_{\max}^*(W)$  is amenable and not quasidiagonal.

*Proof.* Reassociating the semidirect products yields

$$W \cong W_0 \rtimes \mathbb{Z}, \quad W_0 = \text{Cl}(X) \rtimes T_\alpha.$$

For  $n \geq 0$ , write  $\Gamma_n = t^{-n}\Gamma t^n$  for the image of the  $n$ th copy of  $\Gamma$  in  $T_\alpha$ . Since  $[\Gamma : \alpha(\Gamma)] < \infty$ , each  $\Gamma_n$  has finite index in  $\Gamma_{n+1}$ ; every element of  $T_\alpha$  lies in some  $\Gamma_m$ , and conjugation by  $t$  maps  $\Gamma_{m+1}$  onto  $\Gamma_m$ , so  $\Gamma$  is commensurated by  $V$ . Hence the stabilizer in  $\Gamma_n$  of every point of  $X = V/\Gamma$  has finite index, and every  $\Gamma_n$ -orbit in  $X$  is finite.

A finite subset of  $W_0$  involves only finitely many lamps  $c_x$ , and its  $T_\alpha$ -coordinates lie in one  $\Gamma_n$ . Let  $Y$  be the union of the  $\Gamma_n$ -orbits of these finitely many  $x$ . Then  $Y$  is finite and  $\Gamma_n$ -invariant, and the finite subset is contained in

$$C_Y \rtimes \Gamma_n,$$

where  $C_Y = \langle \varepsilon, c_y : y \in Y \rangle$  is finite by the normal form above. This semidirect product is residually finite. Indeed, an element with nontrivial  $\Gamma_n$ -component survives in a finite quotient of  $\Gamma_n$ . For a nontrivial element of  $C_Y$ , let  $J$  be the kernel of the action  $\Gamma_n \rightarrow \text{Aut}(C_Y)$ . The quotient  $\Gamma_n/J$  is finite, and the element has nontrivial image in the finite group  $C_Y \rtimes (\Gamma_n/J)$ . Therefore every finitely generated subgroup of  $W_0$  is residually finite, hence sofic.

The group  $W_0$  is their directed union, and  $W/W_0 \cong \mathbb{Z}$  is amenable. Soficity passes to directed unions and to extensions with amenable quotient [ES, Theorem 1], so  $W$  is sofic. Proposition 4.3 shows that it is not MF. By Proposition 4.2, applied to the extension  $1 \rightarrow W_0 \rightarrow W \rightarrow \mathbb{Z} \rightarrow 1$ , the canonical trace of  $C_{\max}^*(W_0)$  is quasidiagonal and that of  $C_{\max}^*(W)$  is amenable; by Theorem 4.1 the latter is not quasidiagonal.  $\square$

For a concrete instance, take

$$\bar{\Gamma} = \mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z}) = \left\{ \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix} : A \in \text{SL}_3(\mathbb{Z}), v \in \mathbb{Z}^3 \right\} \leq \text{GL}_4(\mathbb{Z}).$$

It is residually finite, since reduction modulo a suitable integer separates any two distinct integral matrices, and it has property (T) [BHV, Example 1.7.4(i)]. Let

$$D = \text{diag}(2, 2, 2, 1), \quad \alpha(g) = DgD^{-1},$$

so that  $\alpha(v, A) = (2v, A)$ . This is injective, and its image consists of the affine matrices whose translation coordinates are all even, so  $[\bar{\Gamma} : \alpha(\bar{\Gamma})] = 8$ . Translation  $a$  by the first standard basis vector lies outside  $\alpha(\bar{\Gamma})$ . So the preceding construction applies to  $\bar{\Gamma}$ ,  $\alpha$  and  $a$ .

*Proof of Theorem B.* Apply Proposition 4.4 to  $\bar{\Gamma}$ ,  $\alpha$ , and  $a$ .  $\square$

### 5. A TORSION-FREE FINITELY PRESENTED EXAMPLE

For an acylindrically hyperbolic group  $G$ , choose the generating set  $A$  provided by Hull [Hu, Theorem 3.12]. A subgroup is *suitable* if it acts non-elementarily on the Cayley graph  $\text{Cay}(G, A)$  and normalizes no nontrivial finite subgroup [Hu, Definition 1.4]. We use the following consequence of Hull's small cancellation theorem [Hu, Theorem 7.1(a), (b), and (e)].

**Theorem 5.1** (Hull's small cancellation theorem). *Let  $G$  be acylindrically hyperbolic, let  $N \leq G$  be suitable with respect to  $A$ , and let  $g_1, \dots, g_m \in G$ . Then there is a surjective homomorphism  $\varphi: G \rightarrow Q$  such that  $Q$  is acylindrically hyperbolic,  $\varphi(g_i) \in \varphi(N)$  for all  $i$ , and every element of finite order in  $Q$  is the image of an element of the same order in  $G$ .*

Hull's proof treats  $m = 1$  by passing to  $G/\langle\langle r \rangle\rangle_G$  for one element  $r$  and the general case by induction on  $m$  [Hu, proof of Theorem 7.1], so  $\ker \varphi$  is the normal closure of  $m$  elements and  $Q$  is finitely presented when  $G$  is.

**Lemma 5.2** (saturation). *Let  $G$  be finitely presented, torsion-free, and acylindrically hyperbolic, and let  $N \trianglelefteq G$  be nontrivial. Then there is a surjective homomorphism  $\varphi: G \rightarrow Q$  such that  $Q$  is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, and  $\varphi(N) = Q$ .*

*Proof.* The subgroup  $N$  is normal, and it is infinite because  $G$  is torsion-free and  $N \neq 1$ , so it acts non-elementarily on  $\text{Cay}(G, A)$  by Osin [Os, Lemma 7.1]; since  $G$  is torsion-free,  $N$  normalizes no nontrivial finite subgroup, so  $N$  is suitable. By Hull [Hu, Corollary 5.7 and Lemma 5.8],  $N$  contains two elements  $h_1, h_2$  such that  $N_0 = \langle h_1, h_2 \rangle$  is suitable with respect to a generating set  $A' \supseteq A$  to which Hull's small cancellation theorem again applies. Apply Theorem 5.1 to  $N_0$ , with respect to  $A'$ , with  $g_1, \dots, g_m$  a finite generating set of  $G$ . Then  $Q = \langle \varphi(g_1), \dots, \varphi(g_m) \rangle \leq \varphi(N_0) \leq \varphi(N) \leq Q$ , so  $Q = \varphi(N_0) = \langle \varphi(h_1), \varphi(h_2) \rangle$  and  $\varphi(N) = Q$ . The quotient is torsion-free by Theorem 5.1 and finitely presented because  $\ker \varphi$  is normally generated by  $m$  elements.  $\square$

Fournier-Facio constructs a finitely presented torsion-free group  $G_0$  with property (T), a subgroup  $\Gamma \leq G_0$  with property (T), an element  $t \in G_0$  with  $t\Gamma t^{-1} \leq \Gamma$ , and a subgroup  $J \leq G_0$  isomorphic to a finitely presented infinite simple group, such that  $[\Gamma, J] = 1$  and  $tJt^{-1} \leq \Gamma$  [FF, §2]. The group  $G_0$  is obtained there as a common quotient of two finitely generated acylindrically hyperbolic groups by Hull's theorem [Hu, Corollary 7.4], which allows the quotient to be chosen acylindrically hyperbolic; we take  $G_0$  to be such a quotient. Put  $S = tJt^{-1}$ . Since  $J$  is simple and nonabelian,  $J$  and  $S$  are perfect.

For  $c \in J$  and  $\ell \in S$ , the commutator  $[tct^{-1}, \ell]$  lies in  $\mathfrak{D}_{G_0}(\Gamma)$  by (1), since  $t$  conjugates  $\Gamma$  into itself,  $c$  centralizes  $\Gamma$ , and  $S \leq \Gamma$ . As  $c$  ranges over  $J$ , the element  $tct^{-1}$  ranges over  $S$ , so  $[S, S] \leq \mathfrak{D}_{G_0}(\Gamma)$ , and  $S$  is perfect, so

$$S \leq \mathfrak{D}_{G_0}(\Gamma).$$

*Proof of Theorem C.* Let  $N$  be the normal closure of  $S$  in  $G_0$ ; it is nontrivial because  $S$  is. By Lemma 5.2 applied to  $G_0$  and  $N$ , there is a surjective homomorphism

$\varphi: G_0 \rightarrow Q$  with  $Q$  two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, and  $\varphi(N) = Q$ . The group  $Q$  is infinite because it is acylindrically hyperbolic, and it has property (T) as a quotient of  $G_0$ .

By (2),  $\varphi(S) \leq \mathfrak{D}_Q(\varphi(\Gamma))$ . The subgroup  $\mathfrak{D}_Q(\varphi(\Gamma))$  is normal in  $Q$ , and the normal closure of  $\varphi(S)$  is  $\varphi(N) = Q$ , so

$$\mathfrak{D}_Q(\varphi(\Gamma)) = Q.$$

Both  $Q$  and  $\varphi(\Gamma)$  have property (T), as quotients of  $G_0$  and  $\Gamma$ , respectively. By the last assertion of Theorem 2.8, every homomorphism from  $Q$  to an MF group is trivial. If a quotient  $\bar{Q}$  of  $Q$  is MF, then the quotient map  $Q \rightarrow \bar{Q}$  is trivial, so  $\bar{Q} = 1$ .  $\square$

**Corollary 5.3.** *The algebra  $C_r^*(Q)$  is separable, unital, generated by two unitaries, simple, has a unique tracial state, has stable rank one, is stably finite, and is not MF.*

*Proof.* The group  $Q$  is countable, torsion-free, and acylindrically hyperbolic, so it contains a non-degenerate hyperbolically embedded subgroup [Os, Theorem 1.2] and has no nontrivial finite normal subgroup. Dahmani, Guirardel, and Osin give simplicity and uniqueness of the trace [DGO, Theorem 2.35], and Gerasimova and Osin give density of the invertible elements, which is stable rank one [GO, Theorem 1.1]. The algebra is separable and generated by the canonical unitaries of the two generators; its canonical trace is faithful, so it is stably finite; and it is not MF by Lemma 2.1, since  $Q$  is not MF (Theorem C).  $\square$

#### ORIGIN AND AUTHORSHIP

In early August 2026 the author was using AI models on two questions raised by the nonsofic group of [OAI]: whether hyperlinearity implies soficity, and whether every group is hyperlinear. In one ChatGPT conversation, GPT-5.6 Sol, on “high” thinking mode, worked on weak MF (matricial field) approximations of a Clifford-twisted wreath product as a step toward hyperlinearity. Earlier in that conversation the author had asked whether to pivot to the MF question, and Sol had advised against it. The author gave Sol the existing progress on the hyperlinear problem and sent this prompt:

“Achieve f{MF of the Clifford Kun–Thom extension } $\widetilde{W}$  in such a way that it archives our goals and ends everything, or, possibly, a different breakthrough, leading to achieving the final result. You got this!”

Sol answered with an MF counterexample: every norm-matrix model of the Clifford-twisted wreath product kills the Clifford sign. That was on August 9. Also on August 9, the models proved that the maximal group  $C^*$ -algebra of a strictly compressed Kazhdan group is infinite. Parallel chats, including one in “Pro” mode, and the agents working on the hyperlinear question, did not find the counterexample. By August 12 the models had reduced the obstruction to one commutator in a finitely presented group and had proved in Lean that it is not MF. They wrote the first draft of this paper that evening.

The author prompted the models, set the direction, retrieved the literature, and suggested the MF question. With the models, the author explored several routes to a non-MF group, shaped the current proof, and checked and revised the citations. The models drafted much of the text; the author wrote the rest and edited the whole for

clarity and for the mathematical presentation. GPT-5.6 Sol in Codex, and Claude Fable 5 and Claude Opus 5 in Claude Code, wrote the Lean proofs. The whole exchange is in the repository’s public commit history.<sup>3</sup>

The author consulted experts from August 13, and their comments shaped the revisions. Many models also reviewed drafts; Astra’s reviews shortened the paper and simplified the proof of Theorem 2.7. We recommend Eckhardt’s writeup [Ec]: a short, elementary proof that certain generalized wreath products are not MF.

## ACKNOWLEDGMENTS

The author thanks Francesco Fournier-Facio for early advice and detailed comments on earlier drafts, and Alon Dogon for organizing the ensuing discussion and for comments on the surrounding literature. The author also thanks Tatiana Shulman for clarifying the terminology and historical context of the MF question, Caleb Eckhardt for discussions of MF representations and matrix norms and for generously writing and sharing the expository note [Ec], Rufus Willett for reading the arguments, and Bruce Blackadar for correspondence on the origin of the MF and NF problems.

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<sup>3</sup><https://github.com/SauersML/group-approximation>

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